Presentation guide for the Xe-124 emulsion experiment

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June 2026

Contents

1 Overview

This document explains every step of the work, from the accelerator to the final results, so you can present it clearly. The overall structure is:

1. **NICA accelerator complex** — how the Xe beam is produced and delivered.
2. **Nuclear emulsion stacks** — what they are and how they detect particles.
3. **Irradiation** — the three stacks and their exposure conditions.
4. **Microscope scanning** — Olympus BX63, ImageJ analysis.
5. **Bragg curve and particle classification** — the main experimental results.
6. **SRIM simulation** — computational model and comparison with experiment.
7. **Conclusions** — what we learned.

2 The NICA accelerator complex (Sec. 1 in the Finalreport)

2.1 What is NICA?

NICA (Nuclotron-based Ion Collider fAcility) is a megaproject at JINR in Dubna, Russia. It accelerates heavy ions (like xenon) for:

- Nuclear physics (quark–gluon matter studies)
- Applied research (radiation hardness of electronics, biology)

2.2 Accelerator cascade

The beam goes through several stages:

1. **ESIS/KRION-6T** (ion source): Produces highly charged Xe^{+54} ions. Xenon gas is ionized by an electron beam or electron string, stripping off 54 electrons.
2. **HILAC** (linear accelerator): Accelerates the ions to 3.2 MeV/nucleon. A linear accelerator uses radio-frequency cavities to push ions in a straight line.
3. **Booster synchrotron**: Accelerates to 600 MeV/nucleon. A synchrotron is a ring accelerator where magnetic fields increase as the particles gain energy to keep them on a circular orbit.
4. **Nuclotron**: The main ring (superconducting magnets). Accelerates to the final energy. For this experiment: $\sim$280 MeV/nucleon for Xe-124.

5. **Extraction:** The beam is *slowly extracted* over several hundred milliseconds (a "spill") to the experimental hall.

2.3 Key point for presentation:

280 MeV/nucleon means each nucleon (proton or neutron) in the xenon nucleus has 280 MeV of kinetic energy. For Xe-124 (124 nucleons), the total kinetic energy is $280 \times 124 = 34720$ MeV ≈ 34.7 GeV. At this energy, the ions are relativistic: $\beta = v/c \approx 0.63$.

3 Nuclear emulsion (Sec. 2 in the Finalreport)

3.1 What is nuclear emulsion?

Nuclear emulsion is similar to photographic film but much thicker and with a higher concentration of silver bromide (AgBr) crystals. When a charged particle passes through:

1. It ionises the medium, creating "latent image" centres in AgBr crystals along its path.
2. After chemical development, these crystals are reduced to metallic silver grains — visible as dark spots under a microscope.

3.2 Key properties

- **Sub-micrometre resolution:** Can resolve individual particle tracks.
- **Integrating detector:** No power needed during irradiation — the emulsion records everything passively.
- **3D information:** By scanning multiple plates, we track particles in depth.
- **Response:** The grain size (area of the silver cluster) is proportional to the energy loss dE/dx of the particle. Heavily ionising particles (like the primary Xe) produce larger grain clusters than light fragments.

3.3 Plate structure

Each plate: 60 µm emulsion on 2 mm glass substrate. The glass provides mechanical support. The emulsion is *single sided* (only one sensitive layer per plate).

4 The three stacks (Sec. 3 in the Finalreport)

Three stacks were irradiated in March 2026:

Stack	Plates	Cycles	Entrance plate	Depth range
Stack 01	8	1	1_Plate_09	0–14.5 mm
Stack 02	10	5	1_Plate-0001	0–18.6 mm
Stack 03	10	10	Plate_01_p1	0–18.6 mm

4.1 Why three stacks?

- Different numbers of exposure cycles (1, 5, 10) give different beam fluences — allows checking statistical effects.
- Stack 01 has only 8 plates (shallower), Stacks 02/03 have 10 (deeper to catch more fragments).

4.2 Depth scale

Depth of plate centre i (0-indexed):

$$d_i = i \times (0.06 + 2.0) + 0.03/2 = i \times 2.06 + 0.03 \text{ mm}$$

4.3 Plate orientation

The layer order within each plate alternates across the stack:

Plate 1: E|G Plate 2: G|E Plate 3: E|G Plate 4: G|E ... Plate 10: G|E

This is important because when the beam hits glass first (plates 2, 4, 6, 8, 10), the energy loss profile differs slightly from hitting emulsion first.

5 Microscope scanning (Sec. 5–6 in the Finalreport)

5.1 Olympus BX63

An automated microscope that scans each plate in a raster pattern, taking thousands of images and stitching them into one large mosaic. At 4× magnification:

- Pixel size: 1.717 µm (Stack 01)
- Scans ∼80% of each plate area
- 2–3 hours per plate

5.2 Why 4×?

The Xe beam at 280 MeV/nucleon produces very large grain clusters (high dE/dx), so even low magnification is sufficient. The trade-off:

- Lower magnification = larger field of view = faster scanning
- Higher magnification resolves individual grains but takes much longer

Comparison images at 4×, 20×, 40× confirm that 4× is adequate.

6 ImageJ analysis pipeline (Sec. 6–7 in the Finalreport)

6.1 Processing steps for each plate

1. **Load** the stitched TIFF mosaic
2. **Set scale:** Calibrate pixels to micrometres (Stack 01: 1.717 µm/px; Stacks 02/03: pre-calibrated)
3. **Enhance contrast:** Normalise the histogram
4. **Threshold:** Select dark pixels (grains) — threshold 0–75 on 8-bit greyscale
5. **Convert to mask:** Binary image
6. **Analyze particles:** Detect connected pixel clusters with size 10–1200 px^2
7. **Export:** Save results CSV with area, perimeter, circularity, position, etc.

6.2 What we measure for each grain cluster

- **Area** — proportional to dE/dx of the parent particle
- **Circularity** ($4\pi A/P^2$) — checks if objects are compact grains or irregular debris
- **Position** (X, Y) — for beam profile analysis

6.3 Scanner calibration note

Stack 01 raw values are in pixel units; the analysis script divides Area by 1.717^2 to convert to $\mu\mathrm{m}^2$. Stacks 02/03 are pre-calibrated — raw values are already in micrometres. This is visible in the CSV column names (`Avg_Area_um2`).

7 Bragg curve and results (Sec. 9–11 in the Finalreport)

7.1 Key computed quantities per plate

$$\text{Track density:} \quad \rho_i = \frac{N_i}{A_i} \quad \text{(grains per mm}^2\text{)} \tag{1}$$

$$\text{Mean grain area:} \quad \bar{a}_i = \frac{1}{N_i}\sum_{j=1}^{N_i} a_{ij} \quad (\mu\text{m}^2) \tag{2}$$

$$\text{Energy deposition fraction:} \quad \varepsilon_i = \frac{1}{A_i}\sum_{j=1}^{N_i} a_{ij} \quad \text{(proxy for } dE/dx\text{)} \tag{3}$$

7.2 Bragg peak

The Bragg peak is where the beam stops — maximum energy loss (dE/dx) just before zero velocity. In the data:

- Track density drops by 14–15× at the Bragg peak
- Stack 01: 8–10 mm depth
- Stacks 02/03: 10–12 mm depth
- The drop is sharp, characteristic of heavy-ion stopping

The dE/dx plot (fractional grain area vs depth) shows the characteristic Bragg curve shape: gradual rise, sharp peak, rapid fall.

7.3 Why the difference between stacks?

Stack 01 peaks ∼2 mm earlier than Stacks 02/03. Possible explanations:

1. Slightly different beam energy between irradiation sessions (magnetic field calibration)
2. Different plate orientations (emulsion upstream vs downstream)
3. Different glass thickness in some plates

8 Gaussian Mixture Model (GMM) classification (Sec. 10 in the Finalreport)

8.1 What is GMM?

Gaussian Mixture Model is an unsupervised machine learning algorithm. It assumes the data comes from a mixture of several Gaussian distributions and finds the most likely parameters. Here we fit 3 Gaussians to the $\log_{10}$(grain area) distribution per plate.

8.2 Why $\log_{10}$?

Grain areas span a wide range ($\sim$10–1000+ μm^2), so logarithm makes the distribution easier to model with Gaussians.

8.3 The three classes

To ensure consistent labelling across plates, the fitted means are sorted in ascending order and re-indexed:

1. **Class 0 (smallest grains, $\sim$12–50 μm^2):**
    - Light fragments: protons, alphas, Li, Be, B (Z $\approx$ 1–5)
    - Low dE/dx, so small grain clusters
    - Dominate after the Bragg peak (long-range fragments)
2. **Class 1 (medium grains, $\sim$40–180 μm^2):**
    - Mid-mass fragments (Z $\approx$ 6–30)
    - Produced by abrasion of the projectile nucleus
    - Dominate in the middle plates
3. **Class 2 (largest grains, $\sim$180–700+ μm^2):**
    - Primary ^{124}Xe and heavy residues (Z $\approx$ 30–54)
    - Highest $dE/dx \rightarrow$ largest grains
    - Fraction peaks at the Bragg peak

8.4 Physical picture (abrasion-ablation model)

As the Xe nucleus travels through the emulsion, it undergoes peripheral nuclear collisions:

1. **Abrasion:** The "edges" of the projectile nucleus are sheared off, producing light and mid-mass fragments.
2. **Ablation:** The remaining excited "prefragment" evaporates additional nucleons.

This produces a spectrum of fragments from single nucleons to near-projectile residues, which we see as the three GMM classes.

8.5 Per-cycle normalisation

Comparing stacks with different exposure cycles:

$$\rho_{\text{cycle}} = \frac{\rho}{n_{\text{cycles}}}$$

Results: Stack 01 highest instantaneous intensity (605/mm^2/cycle), Stack 02 lowest (188/mm^2/cycle), Stack 03 in between (248/mm^2/cycle).

9 Beam profile analysis (Sec. 10 in the Finalreport)

The lateral distribution of grains in each plate reveals the beam shape. 13 diagnostic plots are generated per plate (example shown for Stack 02, Plate 1):

1. **Area distribution** (histogram of grain areas)
2. **Feret diameter distribution** (effective diameter)
3. **Circularity/Roundness** (grain shape quality)
4. **2D density heatmap** (spatial distribution)
5. **2D scatter plot** (every grain as a dot)
6. **Radial density profile** (Gaussian-like fall-off)
7. **Differential flux** (dN/dA vs area)
8. **Cumulative count** (% vs area)
9. **Local fluence map** (10×10 grid counts)
10. **Horizontal slit profile** (X projection)
11. **Vertical slit profile** (Y projection)
12. **XY profile comparison** (roundness check)

9.1 Key findings

- Beam spot is approximately Gaussian, FWHM ~10–15 mm
- Round and symmetric (X and Y profiles match)
- No hot spots, shadows, or edge effects
- Consistent across all three stacks

10 SRIM simulation (Sec. 4 in the Finalreport)

SRIM-2013 was used to simulate the stopping of ^{124}Xe in a 10-plate emulsion–glass stack at the correct experimental energy (280 MeV/nucleon).

10.1 Setup

- 10 plates: 60 µm Ilford G_5 emulsion + 2 mm glass Soda_Lime (total 20.6 mm)
- Emulsion composition: Ag (49%), Br (36%), C (7%), O (7%), H (1%) by weight, density 3.907 g cm^{-3}
- Glass: O (60%), Si (25%), Na (10%), Ca (3%), Mg (1%), Al (1%) by weight, density 2.330 g cm^{-3}

10.2 Key results

- **Range:** 11.55 mm (ion stops at end of plate 6)
- **Bragg peak:** 5505 MeV/mm at 8.30 mm depth
- **Peak-to-surface ratio:** ∼2.0

10.3 Energy budget

Most of the beam energy is deposited in plates 2–5: Plate 1: 4669 MeV, Plate 2: 5183 MeV, Plate 3: 5987 MeV, Plate 4: 7585 MeV, Plate 5: 8467 MeV (Bragg peak), Plate 6: 2831 MeV (stop). Plates 7–10 receive no primary energy.

10.4 Comparison with experiment

The SRIM-predicted Bragg peak at 8–10 mm is consistent with the experimental 10–12 mm for Stack 03. The agreement in the rise slope (plates 1–5) supports SRIM stopping power as a valid model.

11 How to explain the key figures

11.1 Figure: Bragg peak overview (bragg_peak_overview.png)

Four panels:

- **Top-left:** Track density vs depth — shows the sharp drop at the Bragg peak where the beam stops.
- **Top-right:** Energy deposition fraction vs depth — peaks just before the density drop.

- **Bottom-left:** Average grain area vs depth — increases as the beam slows (higher dE/dx), drops after the peak.
- **Bottom-right:** Stack configuration summary.

11.2 Figure: SRIM Bragg curve (srim_10plate_bragg.png)

Two panels:

- **Top:** dE/dx vs depth — red = emulsion, blue = glass. Shows the Bragg peak at 8.30 mm (5505 MeV/mm).
- **Bottom:** Remaining kinetic energy vs depth — the ion stops at 11.55 mm (end of plate 6).
- Dashed vertical lines mark the plate boundaries.

11.3 Figure: SRIM vs experiment (srim_vs_exp.png)

Overlay of normalised SRIM-simulated Bragg curve with experimental data from Stack 03 (best statistics). Both curves are normalised to their maxima. The agreement in the rise slope (plates 1–5) is good.

11.4 Figure: dE/dx (bragg_dedx.png)

Three panels, one per stack. The y-axis is the "energy deposition fraction" (grain area / scan area), which is a proxy for dE/dx. Point out:

- Stack 01 peak at 8–10 mm
- Stacks 02/03 peak at 10–12 mm
- The characteristic Bragg curve shape

11.5 Figure: GMM classes

Shows how the three particle populations evolve with depth. Explain:

- Class 0 (light fragments) dominates AFTER the Bragg peak — long-range survivors
- Class 2 (heavy/primary) peaks AT the Bragg peak
- Class 1 (mid-mass) dominates mid-stack
- This follows the abrasion-ablation model

12 Key numbers to remember

Quantity	Value	Notes
Ion	$^{124}Xe^{+54}$	
Energy per nucleon	280 MeV/nucleon	$\beta \approx 0.63$
Total kinetic energy	34.72 GeV	280 × 124
Emulsion thickness	60 µm	
Glass thickness	2 mm	
Stack depth	20.6 mm	10 plates
Bragg peak (exp, S01)	8–10 mm	
Bragg peak (exp, S02/03)	10–12 mm	
Bragg peak (SRIM)	8.30 mm	5505 MeV/mm
Range (SRIM)	11.55 mm	End of plate 6
Track density drop	14–15×	At Bragg peak
GMM classes	3	Sorted by mean area
Per-cycle fluence	188–605 mm^{-2}	Varies by stack
Beam FWHM	10–15 mm	Gaussian profile

13 Potential presentation questions and answers

Q: Why did you use nuclear emulsion instead of an electronic detector?

A: Emulsion provides sub-micrometre spatial resolution and integrates passively over the entire exposure. It's ideal for measuring ranges and fragmentation of heavy ions at these energies. No power or readout is needed during irradiation.

Q: How do you know the grain area is proportional to dE/dx?

A: The developed silver grain size is proportional to the energy deposited in the AgBr crystal along the particle's path. This is a well-established property of nuclear emulsion: higher dE/dx $\rightarrow$ more developed silver $\rightarrow$ larger grain area.

Q: Why did you use 4× magnification?

A: The Xe beam at 280 MeV/nucleon has very high dE/dx, producing large grain clusters even at low magnification. We verified this by comparing 4×, 20×, and 40× images. 4× allows scanning an entire plate in 2–3 hours instead of days.

Q: Why three GMM classes?

A: Physically, we expect three populations: light fragments (Z 1–5), mid-mass fragments (Z 6–30), and heavy residues/primary beam (Z 30–54). The GMM with 3 components finds these naturally. With 2 components we lose the mid-mass class; with 4 the extra component splits one of the physical classes.

Q: Why does the Bragg peak position differ between stacks?

A: Most likely the beam energy varied slightly between the three irradiation sessions (magnetic field uncertainty). The plate orientation (emulsion vs glass facing the beam) and possible glass thickness variations also contribute.

Q: What is the alternating plate orientation?

A: The plates in the stack are arranged such that the order of layers alternates: even-numbered plates have emulsion then glass (E|G), odd-numbered plates have glass then emulsion (G|E). This matches the experimental stack configuration.

Q: What is the practical application of this work?

A: The characterised Xe beam can be used for:

- Radiation hardness testing of electronics (simulating space radiation environment)
- Benchmarking treatment planning codes for hadron therapy
- Nuclear fragmentation studies for space radiation shielding
- Calibrating nuclear emulsion for future NICA experiments

14 Common mistakes to avoid in presentations

1. **Don't say "SRIM range was 16 µm."** The new SRIM run at 280 MeV/nucleon gives range 11.55 mm, which is consistent with the experiment. Always specify which SRIM run you are referring to.
2. **Don't confuse total energy with energy per nucleon.** 280 MeV/nucleon $\times$ 124 = 34.7 GeV total. Always specify which one you mean.
3. **Don't say "dE/dx" when showing grain area.** The grain area is a *proxy* for dE/dx, proportional to energy loss but not in absolute units (MeV/mm). Say "energy deposition fraction, proportional to dE/dx."
4. **Pixel vs micrometre:** Stack 01 raw values are in pixel units; Stacks 02/03 in micrometres. The analysis converts Stack 01 by dividing by 1.717^2.
5. **Stack numbering:** Stack 01 has 8 plates, Stacks 02/03 have 10. The plate naming conventions differ between stacks.